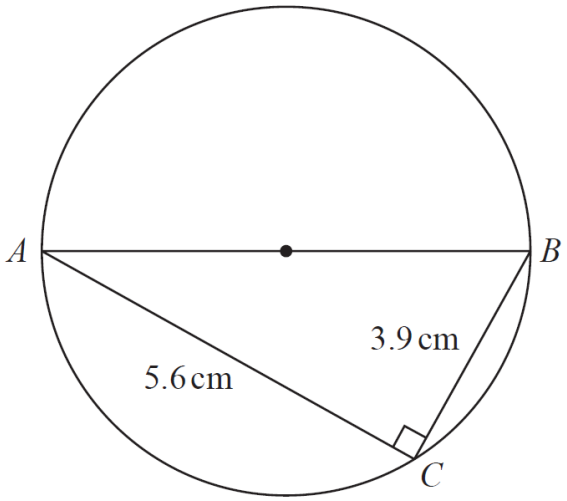
8 A, B and C are points on a circle with diameter AB.
ABC is a right-angled triangle.



Calculate the area of the circle.
Give your answer correct to 3 significant figures.

$AB^2 = 5.6^2 + 3.9^2 = 46.57$

1 mark
Use Pythagoras.

$AB = \sqrt{46.57} = 6.824221567...$ cm

1 mark
Find diameter.

Radius $= 6.824221567... \div 2$

Area $= \pi r^2 = \pi \times 46.57 \div 4$

1 mark
Use half as radius.

$= 36.57599247... = 36.6$ cm^2 (3 s.f.)

1 mark
36.6 cm^2.

…….. cm^2

(Total for Question 8 is 4 marks)